👩‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 9 · STANDARD 6.AT.11

Analyze Graphs of Relationships Between Two Variables

Lesson 9-2 · Teacher Copy

Name: _____ **Date:** _____ **Class:** _____

My Notes

Level 2 Standard

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can use a table and a graph to determine and analyze the relationship between two variable quantities.

Language Objective: I can describe what a point on a graph represents using the words ordered pair, axis, and coordinates.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

graph | table of values | ordered pair | axis | coordinates | Rate of change | Origin

1 A picture on the coordinate plane showing how two quantities relate is a _____.

2. A list of matched pairs that shows a relationship in numbers is a _____ _____ _____ .

3. Two numbers written (x, y) that name one point are an _____ _____ .

4. Each number line framing a graph, labeled with a quantity, is an _____ .

5. The numbers in an ordered pair that tell how far to move are its _____ .

6. How much y changes when x goes up by one is the _____ .

7. The point (0, 0) where the axes meet is the _____ .

Write About the Math

What observations can you make about the data using the graph?
¿Qué observaciones puedes hacer sobre los datos usando la gráfica?

Say your answer to a partner first. Then write it, using at least two of these words:

- ☐ **graph** *gráfica*
- ☐ **table of values** *tabla de valores*
- ☐ **ordered pair** *par ordenado*
- ☐ **axis** *eje*
- ☐ **coordinates** *coordenadas*

Model: *The number of tickets is the independent variable, and the independent variable always comes first in an ordered pair, so the point is (3, 135), not (135, 3).* — A strong answer names tickets as independent and explains that independent always goes first, then checks the arithmetic (3 × 45 = 135). A weak answer just says 'because that's the order' without connecting it to independent/dependent.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **___ is the independent variable, while ___ is the dependent variable. As the years go by, the interest earned ___, because each year adds $___ to the total.**
- **My answer is ___ because ___.**
  Mi respuesta es ___ porque ___.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ___.**
  Mi afirmación es ___.
- **My evidence is ___, so ___.**
  Mi evidencia es ___, entonces ___.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** graph
2. **Notes 2:** table of values
3. **Notes 3:** ordered pair
4. **Notes 4:** axis
5. **Notes 5:** coordinates
6. **Notes 6:** Rate of change
7. **Notes 7:** Origin

8. **Watch for this mistake:** *Reversing the coordinates of a point — writing (cost, tickets) instead of (tickets, cost). The independent variable always comes first in the ordered pair.*
9. **Try It 1:** C. 6,000 feet — *The remaining distance is 18,000 − 12,000 = 6,000 feet — which matches the last 5 minutes at 1,200 feet per minute.*
10. **Try It 2:** C. The Sandia Peak line (1,200 ft/min) rises more steeply than the 900 ft/min line — *In one minute, the Sandia tram climbs 1,200 feet on the graph while the other climbs only 900 — a faster rate makes a steeper line.*